

Build report

Statement Crunch, the lender glossary, report 11 and the bureau captures · 2026-09-18 · demo day Tuesday 22 September

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INSTITUTIONS IN THE GLOSSARY

4

REAL STATEMENTS CRUNCHED

7

BUREAU REPORTS CAPTURED

12 / 14

REPORT TYPES NOW REAL IN THE GALLERY

What the crunching engine now does

Every counterparty on a statement is resolved to an M-PESA shortcode, an account reference and a name, then put against the Central Bank of Kenya's register of 250 digital credit providers, its 40 licensed banks and 15 microfinance banks, and SASRA's deposit-taking SACCOs. A row reading Pay Bill Fuliza M-Pesa to 4145907 - MULAR CREDIT LIMITED Acc. 7981717 is now debt service to a licensed provider carrying an account number, not a trip to the shops. Each identification carries its evidence, and a counterparty that looks like a lender but sits on no register is reported as exactly that.

STATEMENT	MONTHS	ROWS	READ	INCOME / MO	COMMITMENTS	LENDERS	CEILING	SCORE
Elizabeth Charo Washindu	6	8,841	100.0%	KES 114,096	KES 52,951	12	nil	515 Poor
BRIAN KIPKOECH	12	6,832	100.0%	KES 350,596	KES 92,875	2	KES 11,500	529 Poor
DAMARIS WANJIRA NDEGWA	12	5,578	100.0%	KES 81,243	KES 17,956	4 +1?	KES 1,500	578 Poor
Emmanuel Kipleting	6	2,893	100.0%	KES 84,301	KES 14,434	3	KES 1,500	600 Fair

Read is our parse against Safaricom's own printed PAID IN total. All four statements read at 100%, which is what makes the headline checkable: anyone holding the statement can add up page one and get our number. On every one of them Safaricom's own summary does not foot — their printed total differs from the sum of their own rows — and the report prints both figures rather than choosing the one that looks better.

The report

Ten pages, on the crunch theatre's own dark surface rather than the bureau documents' paper. The order is answer first, evidence second, method last: the instalment ceiling, the score and the four worst flags are above the fold on page one; the statement's own summary and our reconciliation are on page two; the lenders, their registers and the month-by-month commitment calendar follow; the affordability arithmetic is shown step by step; and the last page prints the settings the whole thing was produced on, so a number can be defended a year later.

PAGE	WHAT IT CARRIES
1 · Cover	Comfortable instalment, average monthly income, monthly surplus, existing commitments, the score dial, debt-service ratio now and if granted, and the four worst findings.
2 · The statement	Safaricom's header and summary reproduced as printed, our parse totalled beside it, the coverage, and where their own arithmetic does not foot.
3 · Income and expenditure	Money in and money out by month with the average-income line drawn on the same axis, the borrowed share inside each bar, and the net position.
4 · What counts as income	The ladder from the printed total to the figure used, the same statement read the other three ways, and the full monthly ledger.
5 · Where the money goes	Categories, largest counterparties, whether this is a trading account, and supplier-shaped spend.
6–7 · Who else they owe	Every credit provider with its register and the evidence, the commitment calendar, unregistered lenders, and debt being serviced off this rail.
8 · What they can afford	Every step of the arithmetic, both routes side by side, and the settings used.
9 · What moved the score	Twelve drivers, each against its own configured maximum.
10 · Method	Source document and its hash, engine version, policy, what differs from the default, where the engine refused to guess, and a QR code that verifies the document.

Settings, and why they are settings

Average monthly income is not a fact. On Elizabeth's statement the four defensible definitions give KES 114,172, KES 53,804, KES 50,666 and KES 57,587 a month — a factor of two. A microfinance lender writing working capital to traders wants the first; a salary-advance lender wants the third; both are right for their own book. The same is true of the instalment ceiling. So all of it is per-member configuration, Geoffrey can change it himself at `/policy`, and every report prints the settings it was produced on.

What is *not* configurable is the evidence: what the statement says, what Safaricom's summary totals, which counterparties are on the CBK register, how many months the declared period covers. A member configures how to read the evidence, never what the evidence is.

The bureau pulls

Seven billed pulls on national ID 30058967, approved as plan item F3, captured verbatim with a SHA-256 over the bytes written. The preview gallery now shows real answers for 12 of the 14 report types; only report 4 (Metropol's own PDF, which cannot be anonymised without altering the bureau's document) and report 22 (refused, E029, not on the contract) remain.

TYPE	REPORT	LATENCY	INDICATIVE	SHA-256
1	Identity Verification	276 ms	KES 10	6cac762145c9f0144655...
2	Delinquency Status	329 ms	KES 10	c9bac942260b56191301...
5	JSON Credit Report	502 ms	KES 60	ed8ec4d8421aeb44d571...
6	Identity Scrub	239 ms	KES 15	6e7835d45e01dab66749...
10	Enhanced Credit Info	447 ms	KES 55	35f1c6dd2c595628f140...
13	Minified Credit Info	246 ms	KES 20	3f522a78dd5445684262...
14	Full JSON Credit Report	515 ms	KES 90	19aa6e875803c10248f9...
Indicative spend			KES 260	

Worth the money for a reason beyond the gallery. A standalone report 1 returns citizenship, clan, ethnic group, occupation, place of birth, registration office and slots for photo, fingerprint and signature — none of which appear in the identity block nested inside report 12. Showing that nested block and calling it Identity Verification understated the product by about two thirds. Also found: within a single report 1 response, date_of_birth and dob disagree by one day.

Decisions taken, and things found the hard way

Real customer statements were moved out of the web root	Three consented M-PESA statements were sitting in public/, which Vercel serves. Their filenames contain the holders' phone numbers, so the URLs were guessable, and the passwords are six digits. They now live in interchange/private/statements/, which is outside the Next.js web root and git-ignored. Nothing was deployed from public/ before the move.
A bank's paybill is no longer read as debt service	The first cut of the lender matcher counted every payment to Equity's 247247, KCB's 522533 and the rest as a loan repayment. A bank's general paybill takes school fees, rent, deposits and loan repayments through one number and the row does not say which. Counting it all as commitment removed KES 23,921 a month of headroom from one borrower — enough to turn an approvable application into a decline on an assumption. Banks and SACCOs are now only debt service when the row names a loan.
Fund managers are excluded, not flagged as unlicensed lenders	"Nabo Capital Ltd" and "Cytonn Money Market Fund" both trip the CAPITAL legal-form pattern, and the engine duly reported a borrower as servicing an unregistered lender when they were in fact saving. Money to a fund is the opposite signal from money to a lender. Fifteen CMA-licensed managers are now registered specifically to be excluded, and that one fix moved Emmanuel's score from Poor to Fair.
The monthly chart and the headline now measure the same thing	The bars were drawn from classified-income rows while the cover said "statement total paid in". The average line floated above every bar and the document could not be checked against itself. Both now use the member's chosen basis, and both tie to Safaricom's own printed totals.
A new console page was rendering without a session	proxy.ts derives its protected list from the rights catalogue, but Next.js requires the route matcher to be a static literal, so the two are maintained separately. /policy was added to one and not the other, and the page rendered to anyone who typed the URL. Fixed, and a build-time assertion now fails the next time the two drift — which is the actual fix.
Four-digit operator codes are enabled on the public deployment	Production refuses access codes unless INTERCHANGE_ALLOW_CODE_LOGIN is 1. It is set, because the requirement is that Geoffrey signs in and configures his own settings and a browser cannot produce the Ed25519 signature the real door expects. Five failures lock the operator for fifteen minutes and there is a per-IP ceiling above that. This should go back to 0 the day after the demo.

Still outstanding

Metropol's written answer (E1)	Bureau Direct stays Micromart-only until Metropol confirm reports may be pulled for other lenders through an intermediary. This is still the largest commercial risk in the plan and nothing here changes it.
The Registry still runs on Supabase in Ireland	B1–B3 are done, so the move to servicesuite-in can start. Until it does, the p95 target of 400 ms is a measurement of distance, not of the fan-out.
The CallBox superuser password	Still in plain text in a repository, for a Postgres port that answers from the public internet. The Registry moving onto that machine inherits the problem. Rotating it needs coordinating with whoever runs CallBox.
ODPC registration number	Not supplied, so it does not print on any document. A plausible-looking number on a regulated report would be a fabrication.
SVG and knockout artwork	Three PNGs were supplied and everything built so far is derived from them. An SVG master and a white lockup are needed before signage, merchandise or any dark deck.
Statement Crunch in the live portal	The engine, the policy layer and the document are done and proven on four real statements. Wiring the upload route, the metering and the theatre into the member portal is the next block of work.

Signing in on Tuesday

WHO	ROLE	SCOPE	OPENS
Birgen Krosovic	Super admin	Platform	Everything, including governance and every member's settings.
Geoffrey Njane	Member admin	KE/LENDER/3002	Micromart's own surfaces, and /policy to change how his statements are read.

Access codes were sent separately and are not written in this document. Turn `INTERCHANGE_ALLOW_CODE_LOGIN` back to `0` once the demo is over — four digits on a public URL is survivable for a week because of the lockout, and is not a permanent arrangement.